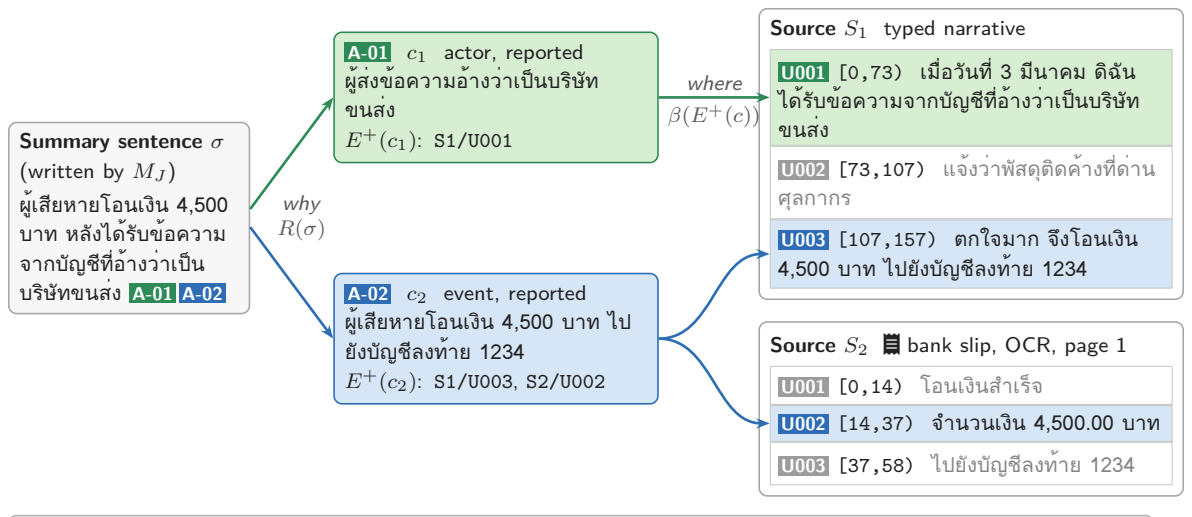

Binding β on every where-arrow: expand the alias with the recorded h_S (`S1/U003` → `S1:U003-80a1d348f2908e99`), then check in order `unknown_source`, `malformed_id`, `cross_source`, `stale_id`, `unknown_unit`, `empty_unit`; plus `duplicate_id`.

✅ pass: reproduce $S[a{:}b]$ with its page ❌ fail: shown as an invalid pointer, never repaired

σ is **bound** when every claim in $R(\sigma)$ has resolved support (otherwise **mixed**, **unbound** or **no_claim**).